

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$a_1=1, a_2=a_3=a_4=0$ Satisfies Column 1

$b_4=1, b_1=b_2=b_3=0$ satisfies Column 2

$$\therefore D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Let $C = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix}$ $CA = \begin{bmatrix} C \begin{bmatrix} 1 \\ 0 \end{bmatrix} & C \begin{bmatrix} 1 \\ 1 \end{bmatrix} & C \begin{bmatrix} 1 \\ 1 \end{bmatrix} & C \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix}$ (By defn. of matrix multiplication)

$$C \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + 0 \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (\text{By Defn. of } Ax)$$

$$= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad (\text{By scalar multiplication and vector addition})$$

$$C \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + 1 \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (\text{By Defn. of } Ax)$$

$$= \begin{bmatrix} x_1+y_1 \\ x_2+y_2 \\ x_3+y_3 \\ x_4+y_4 \end{bmatrix} \quad (\text{By scalar multiplication and vector addition})$$

$$C \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + 1 \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (\text{By Defn. of } Ax)$$

$$= \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (\text{By scalar multiplication and vector addition})$$

$$\therefore CA = \begin{bmatrix} x_1 & x_1+y_1 & x_1+y_1 & y_1 \\ x_2 & x_2+y_2 & x_2+y_2 & y_2 \\ x_3 & x_3+y_3 & x_3+y_3 & y_3 \\ x_4 & x_4+y_4 & x_4+y_4 & y_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$x_2+y_2=1$ and $x_2+y_2=0$ are inconsistent. $\therefore C$ does not exist. $\square$